EXPERIMENT-1

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USER INTERFACE AND DESIGN

Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

AIM:

To design a user interface that helps users recall visual elements such as icons or text chunks and to evaluate the effect of chunking on user memory.

FRAME 1:

INSTRUCTION PAGE:

This page represents the Instruction Screen of the Memory Recall Task application. The primary purpose of this screen is to clearly explain the task flow and rules before the user begins the activity.

Chunking Analysis of the Instruction Page:

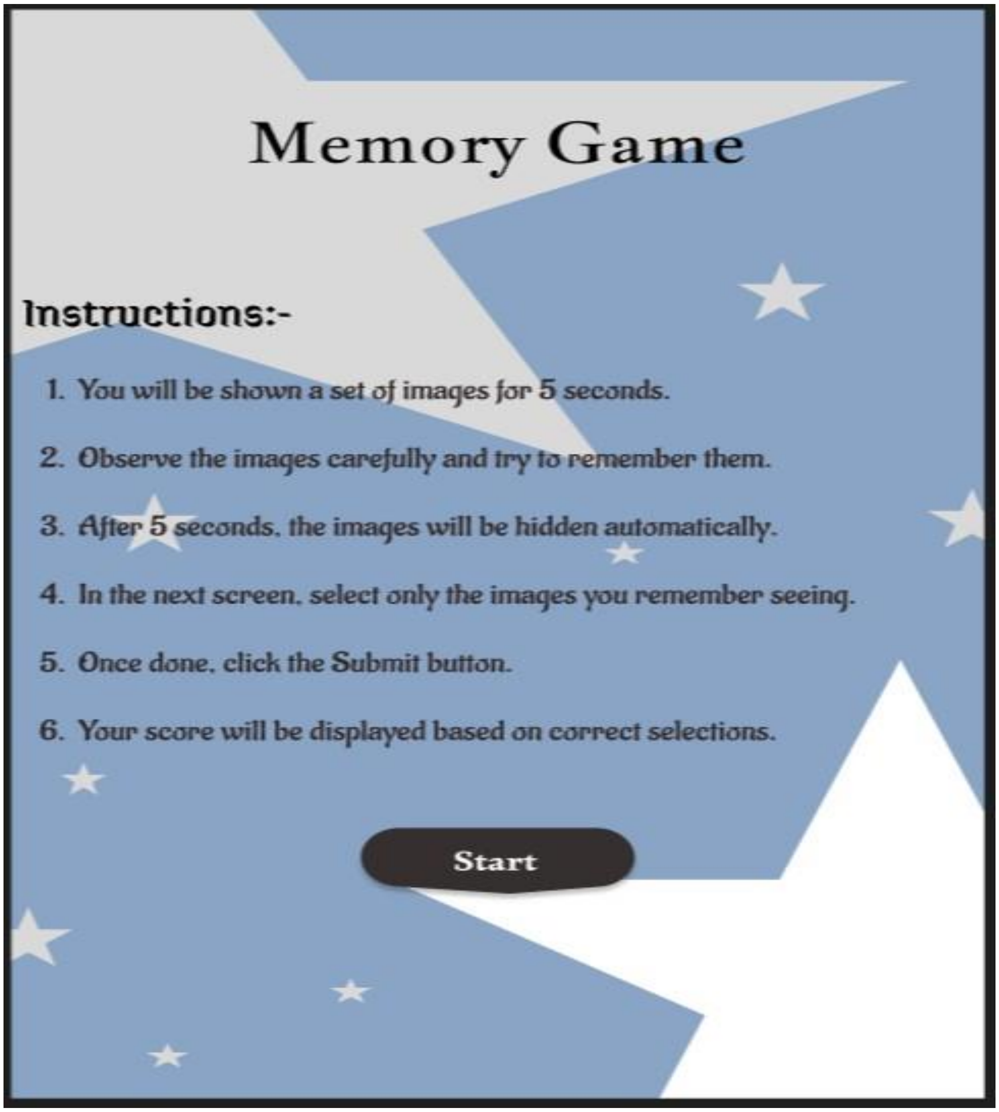
1. Clear and Sequential Instructions: The instructions are presented step-by-step, ensuring users understand the task flow without confusion.

2. Logical Grouping of Information: The steps are grouped into observation, memorization, and recall phases, helping users mentally prepare for each stage.

3. Visual Hierarchy: Bold headings, spacing, and bullet points improve readability and reduce cognitive load.

4. Time Constraint Awareness: Mentioning the limited time prepares users to focus during the recall phase.

5. Simplicity and Clarity: Short sentences and active voice make the instructions easy to understand.



FRAME 2:

MEMORY RECALL PAGE :

This page represents the Memory Recall or Chunking Phase of the task. Users are shown multiple visual elements for a duration of five seconds, during which they must observe and memorize the items displayed.

Analysis of the Memory Recall Page:

1. Purpose of the Page: This page acts as the memory encoding stage where users form mental chunks.

2. Countdown Timer: The five-second timer creates urgency and forces focused attention.

3. Grid Layout of Icons: Icons are arranged systematically to help users identify patterns and categories.

4. Chunking Strategy: Users naturally group similar icons based on color, shape, or theme.

5. Cognitive Benefits: Chunking reduces memory load and improves short-term recall efficiency.

FRAME 3:

SELECTION SCREEN:

This page represents the Selection Screen, where users recall the previously viewed items and select the correct ones from a set of options.

Analysis of the Selection Screen:

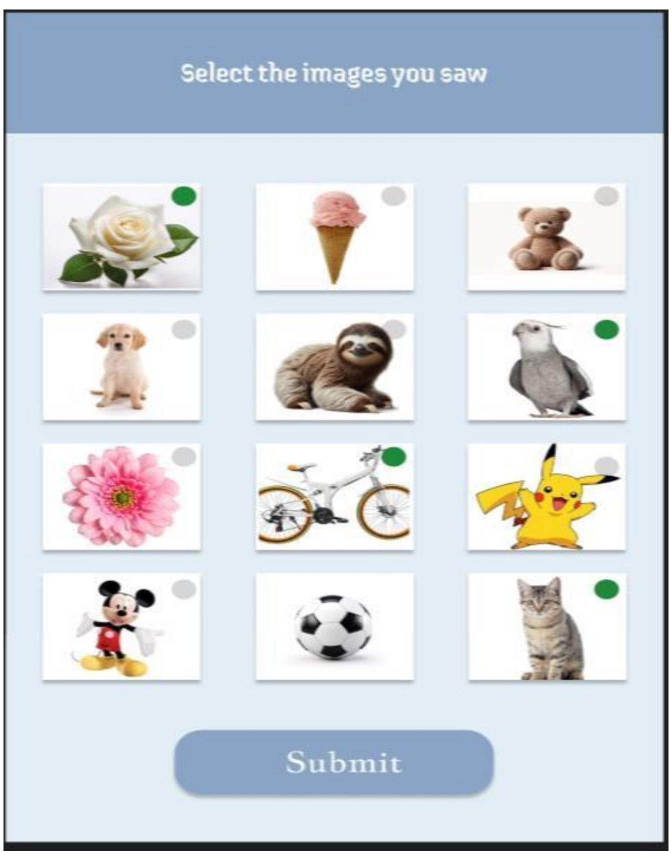
1. Memory Retrieval Phase: This stage evaluates how effectively users recall stored information.

2. Multiple-Choice Icons: Users are shown a mix of correct items and distractors.

3. Use of Distractors: Similar-looking icons test false memory and recall accuracy.

4. User Interaction: Simple selection controls ensure ease of use.

5. UX Benefits: Clear instructions reduce confusion and improve task performance.



FRAME 4:

SCORE SCREEN:

This page displays the final score and feedback based on the user's selections. It helps users understand their performance and decide their next action.

Analysis of the Score Page:

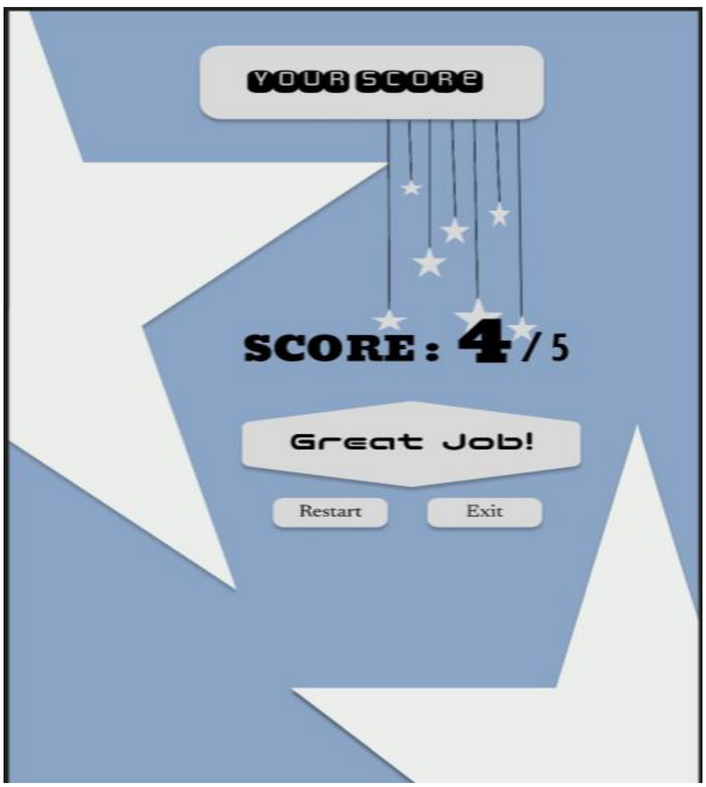
1. Score Display: Shows the number of correct answers clearly.

2. Immediate Feedback: Helps users assess their memory performance.

3. Action Buttons: Restart and Exit options provide user control.

4. Gamification Elements: Visual characters and themes increase engagement.

5. Learning Outcome: Encourages users to improve memory strategies through repetition.



RESULT:

The designed UI successfully demonstrated that users could remember grouped visual elements more effectively than randomly arranged elements. Chunking improved recall accuracy and user memory performance.